

A Level 9709 P1 + P3 练习题

练习题

Q1 [3 marks]

Explain or calculate a result using quadratics in the topic P1–1 Algebra and functions. Show all working.

Q2 [3 marks]

Explain or calculate a result using modulus in the topic P1–1 Algebra and functions. Show all working.

Q3 [3 marks]

Explain or calculate a result using inequalities in the topic P1–1 Algebra and functions. Show all working.

Q4 [5 marks]

Explain or calculate a result using composite and inverse functions in the topic P1–1 Algebra and functions. Show all working.

Q5 [5 marks]

Explain or calculate a result using quadratics in the topic P1–1 Algebra and functions. Show all working.

Q6 [3 marks]

Explain or calculate a result using circles in the topic P1–2 Coordinate geometry. Show all working.

Q7 [3 marks]

Explain or calculate a result using lines in the topic P1–2 Coordinate geometry. Show all working.

Q8 [3 marks]

Explain or calculate a result using intersections in the topic P1–2 Coordinate geometry. Show all working.

Q9 [5 marks]

Explain or calculate a result using circles in the topic P1–2 Coordinate geometry. Show all working.

Q10 [5 marks]

Explain or calculate a result using lines in the topic P1–2 Coordinate geometry. Show all working.

Q11 [3 marks]

Explain or calculate a result using radians in the topic P1–3 Circular measure and trigonometry. Show all working.

Q12 [3 marks]

Explain or calculate a result using exact values in the topic P1–3 Circular measure and trigonometry. Show all working.

Q13 [3 marks]

Explain or calculate a result using identities in the topic P1–3 Circular measure and trigonometry. Show all working.

Q14 [5 marks]

Explain or calculate a result using equations in the topic P1–3 Circular measure and trigonometry. Show all working.

Q15 [5 marks]

Explain or calculate a result using radians in the topic P1–3 Circular measure and trigonometry. Show all working.

Q16 [3 marks]

Explain or calculate a result using AP in the topic P1–4 Series and binomial. Show all working.

Q17 [3 marks]

Explain or calculate a result using GP in the topic P1–4 Series and binomial. Show all working.

Q18 [3 marks]

Explain or calculate a result using sigma notation in the topic P1–4 Series and binomial. Show all working.

Q19 [5 marks]

Explain or calculate a result using binomial expansion in the topic P1–4 Series and binomial. Show all working.

Q20 [5 marks]

Explain or calculate a result using AP in the topic P1–4 Series and binomial. Show all working.

Q21 [3 marks]

Explain or calculate a result using power rule in the topic P1–5 Differentiation. Show all working.

Q22 [3 marks]

Explain or calculate a result using tangent in the topic P1–5 Differentiation. Show all working.

Q23 [3 marks]

Explain or calculate a result using stationary points in the topic P1–5 Differentiation. Show all working.

Q24 [5 marks]

Explain or calculate a result using kinematics in the topic P1–5 Differentiation. Show all working.

Q25 [5 marks]

Explain or calculate a result using power rule in the topic P1–5 Differentiation. Show all working.

Q26 [3 marks]

Explain or calculate a result using reverse differentiation in the topic P1–6 Integration. Show all working.

Q27 [3 marks]

Explain or calculate a result using definite integrals in the topic P1–6 Integration. Show all working.

Q28 [3 marks]

Explain or calculate a result using area in the topic P1–6 Integration. Show all working.

Q29 [5 marks]

Explain or calculate a result using reverse differentiation in the topic P1–6 Integration. Show all working.

Q30 [5 marks]

Explain or calculate a result using definite integrals in the topic P1–6 Integration. Show all working.

Q31 [3 marks]

Explain or calculate a result using partial fractions in the topic P3–1 Advanced algebra and functions. Show all working.

Q32 [3 marks]

Explain or calculate a result using rational functions in the topic P3–1 Advanced algebra and functions. Show all working.

Q33 [3 marks]

Explain or calculate a result using transformations in the topic P3–1 Advanced algebra and functions. Show all working.

Q34 [5 marks]

Explain or calculate a result using partial fractions in the topic P3–1 Advanced algebra and functions. Show all working.

Q35 [5 marks]

Explain or calculate a result using rational functions in the topic P3–1 Advanced algebra and functions. Show all working.

Q36 [3 marks]

Explain or calculate a result using laws in the topic P3–2 Logarithmic and exponential functions. Show all working.

Q37 [3 marks]

Explain or calculate a result using e^x in the topic P3–2 Logarithmic and exponential functions. Show all working.

Q38 [3 marks]

Explain or calculate a result using $\ln x$ in the topic P3–2 Logarithmic and exponential functions. Show all working.

Q39 [5 marks]

Explain or calculate a result using modelling in the topic P3–2 Logarithmic and exponential functions. Show all working.

Q40 [5 marks]

Explain or calculate a result using laws in the topic P3–2 Logarithmic and exponential functions. Show all working.

Q41 [3 marks]

Explain or calculate a result using $\sec/\operatorname{cosec}/\cot$ in the topic P3–3 Advanced trigonometry. Show all working.

Q42 [3 marks]

Explain or calculate a result using compound angles in the topic P3–3 Advanced trigonometry. Show all working.

Q43 [3 marks]

Explain or calculate a result using R-form in the topic P3–3 Advanced trigonometry. Show all working.

Q44 [5 marks]

Explain or calculate a result using identities in the topic P3–3 Advanced trigonometry. Show all working.

Q45 [5 marks]

Explain or calculate a result using $\sec/\operatorname{cosec}/\cot$ in the topic P3–3 Advanced trigonometry. Show all working.

Q46 [3 marks]

Explain or calculate a result using product in the topic P3–4 Differentiation and applications. Show all working.

Q47 [3 marks]

Explain or calculate a result using quotient in the topic P3–4 Differentiation and applications. Show all working.

Q48 [3 marks]

Explain or calculate a result using chain rule in the topic P3–4 Differentiation and applications. Show all working.

Q49 [5 marks]

Explain or calculate a result using implicit in the topic P3–4 Differentiation and applications. Show all working.

Q50 [5 marks]

Explain or calculate a result using parametric in the topic P3–4 Differentiation and applications. Show all working.

Q51 [3 marks]

Explain or calculate a result using substitution in the topic P3–5 Integration and differential equations. Show all working.

Q52 [3 marks]

Explain or calculate a result using by parts in the topic P3–5 Integration and differential equations. Show all working.

Q53 [3 marks]

Explain or calculate a result using partial fractions in the topic P3–5 Integration and differential equations. Show all working.

Q54 [5 marks]

Explain or calculate a result using numerical methods in the topic P3–5 Integration and differential equations. Show all working.

Q55 [5 marks]

Explain or calculate a result using substitution in the topic P3–5 Integration and differential equations. Show all working.

Q56 [3 marks]

Explain or calculate a result using Newton–Raphson in the topic P3–6 Numerical methods and vectors. Show all working.

Q57 [3 marks]

Explain or calculate a result using vector lines in the topic P3–6 Numerical methods and vectors. Show all working.

Q58 [3 marks]

Explain or calculate a result using scalar product in the topic P3–6 Numerical methods and vectors. Show all working.

Q59 [5 marks]

Explain or calculate a result using Newton–Raphson in the topic P3–6 Numerical methods and vectors. Show all working.

Q60 [5 marks]

Explain or calculate a result using vector lines in the topic P3–6 Numerical methods and vectors. Show all working.